

6. **[40 pts]** Consider the following Bayesian binary classification model with a Bernoulli likelihood over i.i.d. input/output observations  $\mathcal{D} = (\mathbf{X}, y)$ :

$$p(y|\mathbf{X}, W, \gamma) = \prod_{i=1}^N a_i^{y_i} (1 - a_i)^{1-y_i}, \quad (1)$$

where  $a_i = \sigma(w^T x_i)$ , with  $\sigma()$  denoting the logistic sigmoid function. The weights  $W$  and their precision  $\lambda$  are assigned prior distributions:

$$p(W|\lambda) = \mathcal{N}(w|0, \lambda^{-1}),$$

$$p(\lambda) = \text{Gam}(\lambda|\alpha_0^\lambda, \beta_0^\lambda)$$

where  $(\alpha_0^\lambda, \beta_0^\lambda)$  are assumed to be known parameters, and Gam denotes the Gamma distribution.

- (a) **[5 pts]** Use Bayes rule to express the posterior over the model parameters  $p(W, \lambda|\mathbf{X}, y)$  as a function of the likelihood and the priors.
- (b) **[15 pts]** Derive the optimization objective for training the model parameters  $(W, \lambda)$  using maximum likelihood estimation (MLE). In other words, derive the expression for  $-\log p(y|\mathbf{X}, W, \lambda)$  using the definition of the Bernoulli probability mass function defined in equation 1.
- (c) **[20 pts]** Derive the optimization objective for training the model parameters  $(W, \lambda)$  using maximum a-posteriori estimation (MAP). To do so, recall that the posterior over the model parameters is proportional to  $p(W, \lambda|\mathbf{X}, y) \propto p(y|\mathbf{X}, W, \lambda)p(W|\lambda)p(\lambda)$ , and use the following expression for computing the log of the Gamma distribution:

$$\log \text{Gam}(c|\alpha, \beta) = (\alpha - 1) \log c - \beta c + \log c.$$

5. **[30 pts]** Provide a short answer to the following questions:

- (a) Briefly describe something that can go wrong if we choose too high of a learning rate for full batch gradient descent. **[5 pts]**

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- (b) Briefly describe something that can go wrong if we choose too low of a learning rate for full batch gradient descent. **[5 pts]**
- (c) Briefly describe the different techniques you know for regularizing deep neural networks. **[10 pts]**
- (d) Briefly explain what is meant by over-fitting. Is it true that if you choose the hyper-parameters (e.g. number of hidden units in a neural network) well, then there will be no over-fitting? Why or why not? (Either YES or NO is acceptable, as long as you justify your answer.) **[10 pts]**

Figure 1: Ignore c,d

1. [15 pts] True or False. Give short reasons for your answers:

- (a) A classifier trained with a lower training rate is less likely to overfit.
- (b) Both linear regression and logistic regression are special cases of neural networks.
- (c) The MLE estimate for the parameters  $(\alpha_1, \alpha_2)$  of the following model can be computed analytically:  $y_i = \log(x_i^{\alpha_1} e^{\alpha_2}) + \epsilon_i$ , (where  $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$  is i.i.d noise).
- (d) If you increase the number of hidden layers in a multi layer perceptron, the classification error of test data always decreases.
- (e) There's no difference between applying the activation function after applying dropout, or applying the activation function before dropout.
- (f) Cross-entropy loss can take on negative values.

**Solution:**

- (a) False. Overfitting isn't related to the training rate. Training rate is determined by the size / complexity of our model and the quantity of data available.
- (b) True. Linear Regression is just an MLP with no hidden layers. Logistic regression is an MLP with no hidden layers with a sigmoid activation on the output layer.
- (c) True. The model is linear in terms of  $(\alpha_1, \alpha_2)$ :  $y_i = \log(x_i^{\alpha_1} e^{\alpha_2}) + \epsilon_i = y_i = \alpha_1 \log(x_i) + \alpha_2 + \epsilon_i$ .
- (d) False. Increasing model complexity may lead to overfitting on the training data and this lead to worse performance on the test set.
- (e) True / False (justification matters): Dropout would make certain neurons to zero. If we consider activation functions that map 0 to 0, then the statement holds true. Otherwise, the output of the neuron would be  $\phi(0)$ , where,  $\phi$  is the activation function considered.
- (f) False. The cross-entropy loss is given by  $-\sum p_i \log(\hat{p}_i)$ . We know that  $\hat{p}_i \in [0, 1] \implies \log(\hat{p}) < 0$ . Also, we know that  $p_i \in \{0, 1\}$ . So the cross-entropy loss must always be positive.

Figure 2: Ignore b,d,e

3. [10 pts] A junior data scientist at the center for disease control (CDC) models the possible spread and containment of a virus using a numerical simulation. He knows that out of a population of  $k$  humans (the number of trials),  $x$  are carriers of the virus. He believes the sample likelihood of the virus in the population, follows a Binomial distribution:

$$L(\gamma|x) = \binom{k}{x} \gamma^x (1-\gamma)^{k-x}, \quad \gamma \in [0, 1], x = 1, 2, \dots, k. \quad (6)$$

You guide him that his parameter of interest is  $\gamma$ , the proportion of infected humans in the entire population. The expectation and variance of the binomial distribution are:

$$\mathbb{E}(x|\gamma, k) = k\gamma, \quad \text{Var}(x|\gamma, k) = k\gamma(1-\gamma). \quad (7)$$

- (a) In a population of 10,000 individuals, 300 were infected. Find the MLE for  $\gamma$ .  
 (b) Find the Fisher information  $I(\gamma)$ . (Hint: Recall that in HW3 you proved that negative expected Hessian of log-likelihood is equal to the Fisher Information Matrix.)

**Solution:** We have that

$$\log(L) = c + x \log \gamma + (k-x) \log(1-\gamma) \quad (8)$$

$$g(\gamma) = \frac{x}{\gamma} + \frac{x-k}{(1-\gamma)} \quad (9)$$

$$H(\gamma) = -\frac{x}{\gamma^2} - \frac{k-x}{(1-\gamma)^2} \quad (10)$$

$$I(\gamma) = -\mathbb{E}(H(\gamma)) = \frac{k}{\gamma} + \frac{k}{(1-\gamma)} = \frac{k}{\gamma(1-\gamma)} \quad (11)$$

Our MLE estimate can be found by setting  $g(\gamma) = 0$

$$(1-\gamma_{\text{MLE}})x + \gamma_{\text{MLE}}(x-k) = 0 \quad (12)$$

$$\implies \gamma_{\text{MLE}} = \frac{x}{k} = 0.03 \quad (13)$$

## 6 Question 6 (20 pts)

Consider the problem of MAP estimation for the mean  $\mu$  of a Gaussian distribution with known standard deviation  $\sigma$ . For the prior distribution, we will use a Gaussian distribution with mean 0 and standard deviation  $\gamma$ .

(a) Determine the function that we need to maximize. You do not need to determine the constant terms explicitly (10 pts).

(b) Determine the optimal value of  $\mu$  by setting the derivative to 0. (You do not need to justify why it is a maximum rather than a minimum.) (10 pts)

(a)

$$\log p(\mu|\mathcal{D}) = \text{const} + \log p(\mu) + \log p(\mathcal{D}|\mu) \quad (1)$$

$$= \text{const} + \log \mathcal{N}(\mu; 0, \gamma) + \sum_{i=1}^N \log \mathcal{N}(x^{(i)}; \mu, \sigma) \quad (2)$$

$$= \text{const} - \frac{1}{2\gamma^2}\mu^2 - \frac{1}{2\sigma^2} \sum_{i=1}^N (x^{(i)} - \mu)^2 \quad (3)$$

(b)

$$\frac{dJ}{d\mu} = -\frac{\mu}{\gamma^2} + \frac{1}{\sigma^2} \sum_{i=1}^N (x^{(i)} - \mu) \quad (4)$$

$$= -\left(\frac{1}{\gamma^2} + \frac{N}{\sigma^2}\right)\mu + \frac{1}{\sigma^2} \sum_{i=1}^N x^{(i)} \quad (5)$$

Letting this to 0, we get:

$$\mu = \frac{\frac{1}{\sigma^2} \sum_{i=1}^N x^{(i)}}{\frac{1}{\gamma^2} + \frac{N}{\sigma^2}} \quad (6)$$

(2) **Linear Classification:** (2 + 2 + 2 + 2 + 7 = 15)

Consider a classification problem where the labels  $Y \in \{-1, 1\}$  and the input features are  $R^d$ . We model it as

$$\hat{y} = \text{sign}(w^T x) \quad (4)$$

- (a) Geometrically what are we doing when we fit this model? Consider drawing a diagram in 2d space to explain it.
- (b) Geometrically, what does  $w$  represent?
- (c) To fit this model, we need a loss function. Ideally, we want a "0-1" loss function:

$$l_{0-1} = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{otherwise} \end{cases} \quad (5)$$

MAX MARKS : 100

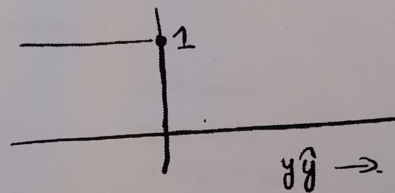


FIGURE 1. The 0-1 loss

Can we use this loss function? If no, give 2 reasons.

- (d) Consider the logistic loss function

$$l_{\log} = \log(1 + e^{-y\hat{y}})$$

Why does the logistic loss look different as compared to the loss we studied in class?

- (e) Derive the weight update rule for  $w$  under the  $l_{\log}$ .